TECHCRUSH AI/ML BOOTCAMP

COHORT 8

GROUP 23

PREDICTING STUDENT EXAM SCORES USING MACHINE LEARNING

CAPSTONE PROJECT REPORT

Team Members:

Obah Edmond Kwame

Adebayo Emmanuel Amos

Favour Ogundele

Akomolafe Abidemi

Orizu Chidiebere

1. INTRODUCTION

Student academic performance can be influenced by different factors. These factors can include the amount of time a student spends studying, attendance, previous academic performance, access to learning resources, motivation, tutoring and other personal or educational conditions.

The purpose of this project is to develop a machine learning model that can predict students' examination scores using these different factors. The problem was treated as a regression problem because the target variable, Exam_Score, is a numerical value.

Two machine learning models were used in the project. These were Linear Regression and Random Forest Regression. The models were trained and evaluated using the same dataset so that their performances could be compared.

The models were evaluated using Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE) and R-squared (R^2). The results showed that Linear Regression performed better than Random Forest on the test data.

The project followed the main stages of a machine learning project, including data acquisition, data cleaning, exploratory data analysis, preprocessing, modelling and evaluation.

2. DATASET DESCRIPTION

The dataset used for this project is called Student Performance Factors. It was obtained from Kaggle and contains information about different factors that may be related to students' examination performance.

The dataset contains 6,607 records and 20 columns. The target variable used for prediction is Exam_Score, while the remaining variables were considered as possible predictors.

Some of the variables in the dataset include Hours_Studied, Attendance, Parental_Involvement, Access_to_Resources, Extracurricular_Activities, Sleep_Hours, Previous_Scores, Motivation_Level, Tutoring_Sessions, Family_Income, Teacher_Quality, School_Type, Peer_Influence, Physical_Activity, Learning_Disabilities, Parental_Education_Level, Distance_from_Home and Gender.

The target variable, Exam_Score, represents the examination score that the model was expected to predict.

The dataset was useful for this project because it contains both numerical and categorical variables. This made it possible to apply different data preparation and preprocessing techniques before training the machine learning models.

3. DATA CLEANING

Before training the machine learning models, the dataset was examined to identify possible data quality problems.

The original dataset contained 6,607 rows and 20 columns. The data was checked for duplicate records and no duplicate rows were found.

3.1 Missing Values

The missing-value analysis showed that three columns contained missing values. These were Teacher_Quality, Parental_Education_Level and Distance_from_Home.

The number of missing values was as follows:

Teacher Quality	78
Parental Education	90
Distance From Home	67

The missing values had to be handled because incomplete data could cause problems during the machine learning process.

The missing categorical values were handled during the cleaning process rather than removing all rows containing missing values. This allowed the project to retain more of the available observations.

After the cleaning process, the dataset was checked again to make sure that the missing-value issue had been addressed.

3.2 Numerical Variable Inspection

The numerical variables identified in the dataset were Hours_Studied, Attendance, Sleep_Hours, Previous_Scores, Tutoring_Sessions, Physical_Activity and Exam_Score.

The minimum and maximum values of these variables were examined to understand their ranges and identify unusual observations.

For example, Hours_Studied had a minimum value of 1 and a maximum value of 44. Attendance ranged from 60 to 100, while Previous_Scores ranged from 50 to 100.

Exam_Score had a minimum value of 55 and a maximum value of 101.

3.3 Outlier Inspection

The Interquartile Range method was used to identify potential outliers in the numerical variables.

The analysis showed potential outliers in variables such as Hours_Studied, Tutoring_Sessions and Exam_Score.

For Hours_Studied, the calculated IQR bounds were 4.00 and 36.00, while the actual values ranged from 1 to 44.

For Tutoring_Sessions, the calculated IQR bounds were -0.50 and 3.50, while the observed values ranged from 0 to 8.

Exam_Score was also inspected for unusually high values. One score of 101 was identified.

The outlier analysis was important because an unusual value does not necessarily mean that the observation is incorrect. Therefore, the values were examined rather than automatically removing every observation identified by the IQR method.

4. EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis was performed to understand the dataset before building the models.

Different visualisations were created to examine numerical relationships, categorical variables and correlations between variables.

4.1 Numerical Variables

Scatter plots were used to examine the relationship between Exam_Score and selected numerical variables.

The variables examined included Hours_Studied, Attendance and Previous_Scores.

The correlation analysis showed that Attendance had the strongest correlation with Exam_Score among the main numerical variables examined.

The correlation values were:

Hours Studied	0.45
Attendance	0.58
Sleep Hours	-0.02
Previous Score	0.18
Tutoring Sessions	0.16
Physical Activity	0.03

Attendance therefore showed a relatively strong positive relationship with examination scores. Hours_Studied also showed a positive relationship with Exam_Score.

The results suggested that some factors could provide useful information for predicting exam scores. However, the relatively weak correlations of some variables also showed that exam performance cannot be explained by only one factor.

4.2 Categorical Variables

Bar plots were created for the categorical variables in the dataset.

These plots were used to understand how the students were distributed across different categories, including parental involvement, access to resources, motivation level, school type and other categorical factors.

The categorical analysis was also important for preprocessing because these variables were stored as categories rather than numerical values. They therefore needed to be converted into a suitable numerical form before being passed to the machine learning models.

5. DATA PREPROCESSING

After the cleaning and exploratory analysis stages, the data was prepared for machine learning.

The dataset contained both numerical and categorical variables. Since the machine learning models require numerical input, the categorical variables had to be encoded.

The numerical variables were also processed as part of the preprocessing stage.

The dataset was divided into training and testing data. The training set contained 5,285 observations, while the test set contained 1,322 observations.

The training data was used to train the models, while the test data was kept separate for evaluating how well the trained models performed on unseen data.

After preprocessing, the feature matrix contained 40 processed features. The increase in the number of features resulted from transforming the categorical variables into numerical features.

The preprocessing steps were fitted using the training data and then applied to the test data.

6. MODELLING

Two different regression algorithms were selected for the project. These were Linear Regression and Random Forest Regression.

Using two models allowed the team to compare different approaches and determine which one performed better on the dataset.

6.1 Linear Regression

Linear Regression was selected as one of the models because it is a simple and commonly used algorithm for predicting numerical values.

It was also useful as a baseline model for the project. The model was trained using the processed training data and then used to predict examination scores in the test set.

6.2 Random Forest Regression

Random Forest Regression was selected as the second model because it can model more complex relationships between input variables and the target.

The Random Forest model was configured with 200 trees and a random state of 42.

The model was trained using the same processed training data and evaluated using the same test data used for Linear Regression.

Using the same test set made it possible to make a fair comparison between the two models.

7. RESULTS AND EVALUATION

The models were evaluated using four regression metrics. These were Mean Absolute Error, Mean Squared Error, Root Mean Squared Error and R-squared.

For MAE, MSE and RMSE, lower values indicate better performance. For R-squared, a higher value indicates better performance.

The results obtained were:

Linear Regression:

MAE	0.452278
MSE	3.255954
RMSE	1.804426
R^2	0.76965

Random Forest Regression:

MAE	1.074164
MSE	40623939
RMSE	2.150335
R^2	0.672875

7.1 Model Comparison

Based on the results, Linear Regression performed better than Random Forest across all four evaluation metrics.

Linear Regression had an MAE of 0.4523, which was lower than the Random Forest MAE of 1.0742. Its MSE and RMSE were also lower.

The R^2 score of Linear Regression was 0.7697, while Random Forest obtained 0.6729.

This means that Linear Regression explained more of the variation in the examination scores in the test data.

Based on these results, Linear Regression was selected as the final model for the project.

The result does not mean that Linear Regression will always perform better than Random Forest. It only shows that, with this particular dataset and preprocessing approach, Linear Regression produced better results as compared to the Random Forest.

8. CHALLENGES AND LESSONS LEARNED

One of the challenges encountered during the project was dealing with missing values. The team had to first identify the columns affected by missing data and then decide how to handle those values without losing any useful observations.

Another challenge was working with both numerical and categorical variables. The categorical variables could not be directly used by the machine learning models, so they had to be transformed during preprocessing.

The outlier analysis was also useful because it showed that some observations were outside the normal IQR range. This helped the team understand that an outlier should not automatically be considered an error.

Another lesson from the project was that a more complex model does not always produce better results. Although Random Forest can model complex relationships, it performed worse than Linear Regression on this dataset.

The project also showed the importance of using several evaluation metrics. Looking at only one metric could give an incomplete picture of model performance.

Working as a group also provided experience in dividing responsibilities, combining different contributions and keeping the project organised using GitHub.

9. CONCLUSION

This project focused on predicting student examination scores using machine learning.

The project followed the main stages of a machine learning workflow, beginning with data acquisition and cleaning, followed by exploratory data analysis, preprocessing, modelling and evaluation.

Two regression models were tested: Linear Regression and Random Forest Regression.

The results showed that Linear Regression performed better on the test data. It achieved an MAE of 0.4523, MSE of 3.2560, RMSE of 1.8044 and R^2 of 0.7697.

Random Forest obtained an MAE of 1.0742, MSE of 4.6239, RMSE of 2.1503 and R^2 of 0.6729.

Based on these results, Linear Regression was selected as the final model.

The project also demonstrated the importance of properly cleaning data, exploring relationships between variables, preparing data before modelling and comparing multiple models.

Overall, the project gave the team practical experience in applying the machine learning process to a real prediction problem and interpreting the results obtained from different models.

10. REFERENCES

Kaggle. (n.d.). Student Performance Factors. Kaggle.

Scikit-learn developers. (n.d.). LinearRegression. Scikit-learn documentation.

Scikit-learn developers. (n.d.). RandomForestRegressor. Scikit-learn documentation.

Scikit-learn developers. (n.d.). Model evaluation: Regression metrics. Scikit-learn documentation.

TechCrush. (2026). Capstone Project Requirements. TechCrush AI/ML Bootcamp Cohort 8.